

ActInf GuestStream #024.1 ~ Stephen Grossberg, "Explainable and Reliable"

GuestStream | Stephen Grossberg | 1:47:17

Hello and welcome everyone. It is June 28th, 2022, and we are here in Acton Flab guest stream number 24.1.

Today we're here with Professor Steven Grossberg, and the agenda will be as follows.

First, Ali will provide a short introduction.

We will then play a 45-minute pre-recorded video, followed by a Q&A.

So, thanks everyone for joining, and Professor Grossberg, really appreciate joining, and I'll pass to Ali for the introduction.

Hello and welcome. I'm Ali. I'm an independent researcher from Iran.

I'm very happy and excited to be here and be able to speak with Professor Grossberg today.

So, I'd like to thank Professor Grossberg for joining us.

Steven Grossberg is the one Professor of Cognitive and Neural Systems and a Professor Emeritus of Mathematics and Statistics,

Psychological and Brain Sciences, and Biomedical Engineering at Boston University.

For more than 50 years, he has led pioneering research in discovering and developing neural design principles for autonomous adaptive intelligence based on biological and machine learning.

His neural network models have been applied to many large-scale problems in engineering and technology, including the design of increasingly autonomous adaptive algorithms and mobile agents.

In fact, this is what Carl Friston says about him.

Whenever you claim to be the first to do this or that in artificial intelligence, it is customary and correct to add, with the exception of Steven Grossberg.

Quite simply, Steven is a living giant and foundational architect of the field.

Professor Grossberg is the recipient of the 2015 Norman Anderson Lifetime Achievement Award of the Society of Experimental Psychologists,

the 2017 Frank Rosenblatt Award of the IEEE Computational Intelligence Society,

and the 2019 Donald O. Hebb Award of the International Neural Network Society.

His latest book, *Conscious Mind, Resonant Brain*, as a combination of his decades-long research, written in a rather non-technical and conversational style,

is published in 2021 by Oxford University Press,

and is the winner of the Association of American Publishers in 2022 Prose Award for the Best Book of the Year in Neuroscience.

Now, I'll pass it to Professor Grossberg, and then we'll continue with the 45-minute pre-recorded lecture.

If you'd like to say hi, otherwise I'll begin the recording video.

I just saw my face frozen on the screen.

Well, I'm delighted to be here, and I hope you find some points of interest in the lecture, and I'll look forward to the Q&A.

Ali has prepared a series of questions that I've thought about

and have some prepared sketched answers,
and then after that, if you're still interested,
I'm happy to do live Q&A about anything related to the topics of the day.
Okay.

On to the main course.

I will play the video.

No?

No.

And you won't hear anything on the live stream.

I'll crop it, and the audio will be coming through fine now.

Hello.

I'm delighted to be able to speak to you today about a topic concerning artificial intelligence,
which, as you know, is very much in the news these days,
and I'll be contrasting two very different approaches to artificial intelligence.

But to do that, I need to pull up my PowerPoint slides and share them with you,
and let me maximize them and minimize my face.

So, my topic today is explainable and reliable AI,
comparing deep learning with adaptive resonance.

This lecture is based on the following article from this year,
which is both open access and on my webpage.

The article summarizes core problems of deep learning,
such as its untrustworthiness because it's unexplainable,
and its unreliability because it experiences catastrophic forgetting.

The article explains how adaptive resonance overcomes these problems,
indeed overcomes 17 problems of deep learning,
and outlines a blueprint for achieving autonomous adaptive intelligence.

The article is part of a Frontiers in Neurobotics special issue about explainable AI,
whose editors wrote, and I quote,

Though deep learning is the main pillar of current AI techniques,
and is ubiquitous in basic science and real-world applications,
it is also flagged by AI researchers for its black box problem.

It is easy to fool,
and it also cannot explain how it makes a prediction or decision.

In other words, deep learning is not trustworthy.

No life or death decision,
such as a medical or financial decision,
can confidently be made based upon a deep learning prediction.

Deep learning uses the backpropagation algorithm

for learning how to predict output vectors in response to input vectors.
Backpropagation was based on perceptron learning principles
that Frank Rosenblatt started to introduce in the 1950s.
It has a complicated history,
which Juergen Schmidhuber beautifully reviewed in an article from this year.
Major contributors include Shunichi Omari,
Paul Wervos, and David Parker.
Perhaps one would say that it reached its modern form
with simulated applications
in Paul's 1974 paper,
before being popularized 12 years later
by Rumelhart, Hinton, and Williams.
Here's a schematic of a backpropagation circuit
were printed from a survey article
by Gail Carpenter of Neural Network Models.
In it, information flows feed forward
from an input stage to an output stage.
Learning is supervised by an external teacher
who on each trial defines a target or desired output.
The teaching signal is the error or mismatch
between the actual and the target outputs.
The teaching signal in level F3
of adaptive weights in level F2
have no network pathway whereby
to reach from F3 to F2 within the algorithm.
So the algorithm uses an artifice
called weight transport,
which physically lifts the weights from here
and moves them there so that they can be used
to control learning.
Well, this is clearly a non-local operation,
as well as being clearly non-biological.
Backpropagation learns through slow learning,
which means that the adaptive weights change
just a little to reduce error on each learning trial.
That requires many trials,
that is to say many repetitions
of the whole database to learn,

possibly hundreds or thousands of trials.

This is to be contrasted with fast learning,
where adaptive weights zero error signals
on each trial,

just as we can learn a face that we see just once
and remember it for a long time.

If Backprop tried to use fast learning,
it would become wildly unstable.

Catastrophic forgetting also occurs in Backprop,
so that during any learning trial,
an unpredictable part of its learned memory
can unexpectedly collapse.

So deep learning,
which is based on Backpropagation,
is thus neither reliable nor trustworthy.

But why is this?

One reason is that all inputs are processed
by a shared set of learned weights.

The algorithm cannot selectively buffer learned weights
that are still predictably useful.

In particular, there's no attention mechanism.

This problem occurs in any learning algorithm
whose shared weight updates
follow the gradient of the error
in response to the current batch of data points
while ignoring past batches.

There have been multiple efforts
to fix backpropagation.

One is to selectively slow learning
on the weights important for learning
by optimizing parameters using the Bayes rule,
as Kirkpatrick et al. suggested a few years ago.

But that assumes an omniscient observer
who can discover and alter
the important weights,
as well as non-local computations,
such as the Bayesian computation.

The same problem occurs with evolutionary algorithms

and diffusion-based neuromodulation
and other approaches to try to fix backprop.
These efforts to overcome catastrophic forgetting
created additional conceptual
and computational problems.

I view them as adding epicycles
to ameliorate a fundamental flaw in the model,
which to me is reminiscent of adding epicycles
to correct problems in the Ptolemaic model
of the solar system.

As we all know, the Copernical model
that we now accept didn't require epicycles.

Perhaps this is why Jeffrey Hinton,
who played a key role in developing
both backprop and deep learning,
said in an Axios interview a few years ago
that, quote,
he's deeply suspicious of backpropagation.

I don't think it's how the brain works.

We clearly don't need all the labeled data.

My view is throw it all away and start over.

I would claim we don't have to start over
because these problems were solved
in the 1970s and 1980s.

In particular, in the first issue
of the journal Neural Networks in 1988,
I had an article that listed 17 problems
of backpropagation that are overcome
by adaptive resonance, and here they are.

With regard to not needing all the labeled data,

I noted in the third item here
that self-organized unsupervised
or supervised learning frees us
from needing labels all the time.

As to slow learning,

I noted that in art,
you can have fast or slow learning.

Indeed, Art can learn to classify

an entire database
using fast learning
on a single learning trial,
as Gail Carpenter and I showed
in the 1980s.

Moreover, Art overcomes
all 17 problems
of backpropagation
without the cycles.

Furthermore,
all the core art predictions
have been supported
by subsequent psychological
and neurobiological data.

Indeed, Art is a principled
biological and technological theory,
unlike backpropagation
and deep learning,
which are just algorithms.

Art has explained data
from hundreds of experiments,
and it's made scores of predictions
that have subsequently received
experimental support.

Well, why has Art been so successful?

There are a number of reasons,
but one of them is
that Art can be derived
from a thought experiment
about a universal problem
in error correction
that I published 40 years ago
in Psychological Review.

The thought experiment asks the question,
how can a coding error be corrected
if no individual cell knows
that one has occurred?

Let me quote from my paper.

The importance of this issue
becomes clear
when we realize
that erroneous cues
can accidentally be incorporated
into a code
when our interactions
with the environment
are simple
and will only become evident
when our environmental expectations
become more demanding.

And even if our code
perfectly matched
a given environment,
we would certainly make errors
as the environment itself fluctuates.

So I was talking about
autonomous local learning
in a changing world.

A purely logical inquiry
into error correction
is translated at every step
of the thought experiment
into processes
learning autonomously
in real time
with only locally computed quantities.

Moreover, the thought experiment
uses familiar environmental facts
about how we learn
as its hypotheses
and odd circuits
naturally emerge,
where these facts are familiar
because they're ubiquitous
environmental constraints
on the evolution of our brains.

and since we're living with them
all the time,
they become familiar.
Because of this universality,
odd circuits may thus
in some form
be embodied
in all future
truly autonomous
adaptive intelligent devices
whether biological
or artificial.
Odd has probably
for this reason
already been used
in many large-scale
engineering
and technological applications.
In fact,
almost immediately
after art was introduced,
it began being used
because it succeeded
in benchmark studies
against machine learning,
backpropagation,
statistical methods,
genetic algorithms,
either getting
much better accuracy
or much faster training speed
or both.
It's also used
in applications
where other algorithms
totally fail,
such as the Boeing company's
part design,

reuse,
and inventory compression application.

That's just one
of many large-scale applications
in engineering
and technology,
some of which can be found
on our tech lab webpage
at bu.edu.

The Boeing parts design
retrieval system
in particular
was used to help design
the Boeing 777.

And to do that,
you needed fast learning
and stable memory
to learn and search
a huge
and continually growing
non-stationary parts inventory
at the time
of this application.

There are already
16 million,
1 million dimensional vectors
that were used
to describe
each of the parts
and you have to be able
to quickly search
the inventory
if you wanted to find
a part to use
in a new plane,
especially if your new design
might have a part
in the inventory

that was similar to it,
finding it
and slightly modifying design
could save millions
of dollars
in fabrication costs.

Satellite remote sensing
is another large-scale application
that art was used for
very soon.

And Gail Carpenter
and her colleagues
took the lead here.

For example,
using a very small number
of pixels
of ground truth
of 17 vegetation classes,
they used art
to automatically complete
these maps
using remote sensing data.

Art did it in a day,
rapidly and automatically.
It gave a confidence map
for each pixel
and the pixels
were 30 meters in scale,
which was small enough
to see roads.

This contrasted
with an AI expert system,
which took a whole year
to do with
and it had to derive
ad hoc rules
from experts.

You had to correct

upwards of a quarter
of a million site labels
and even so,
the pixel size
was an order
of magnitude larger.
Gail went on
with her colleagues
to study information fusion
in remote sensing.
Let's say you have
multiple observers.
Each of them
may be using
different labels.
Their labels
may also be incomplete
or missing
or even incorrect.
And the task
was to derive
consistent knowledge
from potentially
inconsistent data
to automatically
learn and stably
store
one-to-many mappings.
And along the way,
Gail and a colleague
showed how to self-organize
a hierarchy
of cognitive rules,
including confidence
measures
between these different
levels of the hierarchy.
There's been continual work

on art.

Some more recent work
was summarized
in a special issue
of Neural Networks
just in December 2019
that was edited
by Donald Wunsch,
who started
this special issue
with a general overview
of neural network models
that I and my colleagues
developed
and then went on
in a long and detailed
article
with several collaborators
to provide a survey
of adaptive resonance theory
neural network models
for engineering applications
to the present time.

So backpropagation
and deep learning
are a feed-forward
adaptive filter,
but art is more than that.

In fact,
art is an explainable,
self-organizing production
system
in a non-stationary world.

What do these words mean?

Art is self-organizing
because it can
autonomously carry out
arbitrary combinations

of unsupervised
or supervised learning trials
with the world
as its only teacher.
It's a production system
because it uses
hypothesis testing
to discover and learn rules
by a top-down matching process
that focuses attention
on critical feature patterns.

These are the patterns
that predict behavioral success
while suppressing
irrelevant features.

Art is explainable
using both its activities
or short-term memory
STM traces
and its adaptive weights
or long-term memory
LTM traces.

Activation dynamics,
learning dynamics.

Observing the STM traces
in a critical feature pattern
explain what recognition categories
will learn to code
and what features
predict goal-oriented actions.

In particular,
the long-term memory traces
in the fuzzy arc map algorithm
translate into explicit
fuzzy if-then rules
that code what combinations
of critical features
in what numerical ranges

effectively control predictions,
thereby illustrating
one of many examples
where neural networks
can learn rule-based behaviors.

Art includes
a bottom-up adaptive filter
or feed-forward neural network
as I've observed already,
but that's supplemented
by top-down learned expectations
and two types of recurrent
inhibitory feedback interactions
that help to choose
the recognition categories
and the critical features.

Notably,
top-down expectations
use what Gail Carpenter
and I call
the art matching rule
to learn how to focus attention
on critical features
that control predictive success.

The art matching rule
is another way of talking
computationally about the process
of object attention,
how we pay attention
to salient objects
in the world,
and we show how it stabilizes
learning and thereby avoids
catastrophic forgetting.

Remarkably,
and this has been supported
by many data,
the art matching rule

can be realized
by a top-down modulatory
on-center off-surround network.
Well, what does this mean?
Well, let's say
we have bottom-up inputs
from external features
to feature selective cells
that get stored
in short-term memory.
Let's say we activate
a recognition category
which has previously been learned
and it tries to read out
its learned excitatory prototype.
Well, it can't fully do so
because it also reads out
an inhibitory off-surround
that's broader than the prototype.
So this is approximately
one excitatory
against one's inhibitory.
It can only give you
a modulatory on-center.
But if you have both
bottom-up inputs
and the top-down expectation
simultaneously active,
then within the bounds
of the prototype,
if you also have
a bottom-up feature,
you have two excitatory
against one inhibitory
and those features
can be selected,
gain-amplified,
and synchronized

to start focusing attention
on this critical feature pattern
while outlier features,
the ones that aren't
within the prototype,
only have one excitation
against one inhibition,
are suppressed.

And in 1999,
I was able to begin
to understand
how laminar cortical circuits
carry out object attention.

In particular,
layer six of a higher
cortical area
can activate layer six
of a lower cortical area,
either directly
or via layer five.

And then,
it can fold up
the layer four
to modulate
and on-center
and to inhibit
and off-surround.

So attention acts
via a top-down,
modulatory on-center,
off-surround network
via folded feedback
within laminar neocortex.

And this is one example
of the paradigm
of laminar computing
that I introduced,
which asks,

why are all
neocortical circuits
organized in layers?
And how do laminar circuits
give rise
to all kinds
of biological intelligence?
Adaptive resonance
enters the story
because attended feature clusters
reactivate
their bottom-up pathways.
Activated categories
reactivate
their top-down pathways,
closing an excitatory feedback loop
between features
and categories
giving rise
to a feature category resonance
that synchronizes,
amplifies,
and prolongs
system response
between the attended
critical features
and the category
to which they are bound.
And it's this resonance
that triggers fast learning
in the bottom-up
and top-down
adaptive weights,
which is why
I have called the theory
adaptive resonance theory.
Moreover,
I've done a lot of work

since then
showing that
all conscious states
are resonance states
and these feature category
resonances
are one example of that,
one that supports
conscious recognition
of visual objects
and scenes.

There's a lot of data support
for art predictions.

It's well known
that attention
does have an on-center-off
surround
circuit behind it
and that attention
can facilitate
matched
bottom-up signals,
many other data
as well.

So now we can say
more about why art
is explainable
or trustworthy.

In short-term memory,
it's because
the critical feature patterns
determine the attentional focus
that controls
information processing
and you can just
read off what those features are.
In long-term memory,
again,

it's the critical feature patterns
that determine
the adaptive weights
learned by the
bottom-up adaptive filter
and the top-down
learned expectation
so you know also
what these weights
are encoding.

Art's reliable
and avoids catastrophic
forgetting
because outlier features
that are not
in the critical feature pattern
are suppressed
so that only
the predictive features
are processed
and coded.

Art's a production system
because it carries out
a kind of hypothesis testing
and this is nicely illustrated
in the simplest
art model
called Art1
that Gail Kompater
and I
published in 1987.

Art1 has an attentional
system
that does all the category
learning
and the expectation learning
and the paying attention
that interacts

with an orienting system
which is activated
when there are big enough
matches in the attentional system
and thereby drives
reset and search
for novel
or better matching categories.

Here's a schematic
of the art
hypothesis testing
and learning cycle.

So let's say
you have a bottom-up
feature pattern
coming in.

There may be many,
many active
bottom-up features
but I'll draw just
one arrow here
for simplicity
but that vector
of input features
can activate
a distributed pattern
of feature detector cells.

Some may be very active,
some not so active,
some not active
at all.

And as this is happening,
each of these
active pathways
is trying to
turn on
the orienting system
so there might be

quite a few inputs
converging here
but as the features
are activated,
each of them
tries to inhibit
the orienting system
and there are as many features
as there are inputs
so this excitation
inhibition of balance
keeping the orienting system
quiet
as the feature pattern
goes to the adaptive filter
and chooses
a category.

That category reads out
a learned top-down expectation
that obeys the art matching rule
which can suppress
some mismatched features
thereby reducing
the amount of inhibition
on the orienting system
and raising the question
when you have
too little
inhibition
and too much
excitation
how big a mismatch
will activate
the orienting system
and cause reset
and that
ratio is determined
by what's called

vigilance
which I'll say
more about soon.
But if you don't
have enough inhibition
then the orienting system
gets activated
it equally activates
all the cells
in the category layer
because
it doesn't know
which cell
may be active or not
so it causes
a novelty
sensitive
non-specific
burst of arousal
novel events
are arousing
thereby selectively
shutting off
the active category
eliminating
its top-down
expectation
and unmasking
the original feature
pattern
which can again
go through
the adaptive filter
however now
this
previously
disconfirmed
category

remains off
and
the category
level is
renormalized
so it responds
to the same
input pattern
with a new
category
and you go
through this cycle
of resonance
and reset
until you get
a good enough
match
to either
learn a new
category
or select
a previously
learned category
and it's a theorem
that as categories
are learned
through this matching
process
search automatically
disengages
leading to direct access
without search
to the globally
best matching category
explaining for example
how we can quickly
recognize familiar
objects

like your mother's
face
even if as we get
older we store
enormous numbers
of additional
memories
so you don't have to
search your whole
repertoire
when you see mom
you get direct access
and quickly say
hi mom
there's a lot of
support for the
hypothesis testing
cycle
one source
of support
is from
ventral aid
potentials
also called
human scalp
potentials
which shows
correlated sequences
of three different
evoked potentials
during oddball
learning tests
an experiment
that John
Paul Banquet
and I
reported in the 80s
where you'll get

a P120
for a mismatch
an N200
for the arousal
activated by the
orning system
and a P300
for the short
memory reset
of the category
layer
thereby
supporting
the processing
stages
of the search
cycle
there was also
physiological data
from infratemporal
cortex
where categories
are learned
early on
from the lab
of Bob Desimone
who showed
an active
matching process
that's reset
between trials
during
this kind
of event
there's also
classical data
about hippocampal
mismatch dynamics

it's known
that novelty
potentials
subside
as learning
proceeds
from numerous
experiments
this is
as the orienting
system is disengaged
and there's
more recent data
using multiple
electrode studies
from the lab
of Earl Miller
from prefrontal
cortex
and simultaneous
recordings
in a hippocampus
and they show
there's rapid
object associative
learning
may occur
in prefrontal
cortex
which is a
projection
of infratemporal
cortex
one of the stages
of category
learning
while the hippocampus
may guide

neocortical
plasticity
for
by signaling
success
or failure
well this is
just what happens
when the
attentional system
interacts
with the
orienting
system
there's also
complementary
computing
in art
in particular
the attentional
and orienting
system
was a
complementary
as manifested
by the fact
that two
event-related
potentials
are complementary
processing
negativity
and M200
processing
negativity
is activated
when there's
a top-down

match
in the
attentional
system
the M200
as I just
noted
is activated
when there's
a mismatch
that activates
the orienting
system
and you can
just look
across
these four
rows
and see
that these
two
kinds of
ERP
potentials
are manifestly
complementary
as
the
illustrator
of the
complementarity
of the
attentional
and orienting
systems
so this
leads us
to discuss

another
paradigm
introduced
which I
call
complementary
computing
that asks
what is
the nature
of brain
specializations
complementary
computing
introduces
new
principles
of uncertainty
and complementarity
that clarify
why there
are multiple
parallel
processing
streams
with
multiple
processing
stages
in our
brains
and a
beautiful
example
of that
is this
famous
image

of the
macro
circuit
of the
visual
system
from
David
Van
Essen
and his
colleagues
where you can
see these
multiple
parallel
processing
streams
and the
multiple
stages
needed to
achieve
what I call
hierarchical
resolution
uncertainty
but what are
complementary
properties
they're
analogies
like a
key
fits into
a lock
or puzzle
pieces

fitting
together
in words
computing
one set
of properties
at a
processing
stage
prevents
that stage
from computing
a complementary
set of
properties
these
complementary
parallel
processing
streams
are
balanced
against
one
another
it's a
very
yin
yang
kind
of
situation
and
interactions
between
the
streams
overcome

their
complementary
weaknesses
in fact
there are
many
complementary
processes
that are
known in
the brain
that have
been
modeled
here is
just
five of
them
there are
many
more
so this
is a
basic
principle
of
brain
organization
so in
summary
so far
back
propagation
and
deep
learning
do
not

have
thought
to
memory
activation
patterns
including
critical
feature
patterns
so they
can't pay
attention
indeed
they don't
have
any
fast
information
processing
nor do
they have
long-term
memory
top-down
learned
expectations
so they
can't
carry out
hypothesis
testing
using
interactions
short-term
and long-term
memory
traces

indeed
there's
no
neural
architecture
there's
just
an
algorithm
in
really
great
contrast
with
complementary
computing
which
discusses
the global
organization
of our
brains
from the
very start
it was
shown
how easy
it is
to get
catastrophic
getting
and
Carpenter
and I
showed
it
in
art

when
we
would
shut
down
the art
matching
rule
then
we
demonstrated
you could
get
catastrophic
forgetting
if you
had
just
four
input
vectors
a
b
c
d
presented
in the
order
a b
c
a d
a b
c
a d
and so
on
if they
obeyed

very
simple
subset
relationships
and here's
a computer
simulation
of that
here you
don't have
the art
matching
rule
here's
a b
c
a d
a b
c
a d
and you
see a
is coded
by category
one here
by category
two here
by category
one here
two here
it never
settles
down
but as
soon as
you impose
the art
matching

rule
learning is
complete by
the second
trial
and after
that point
you get
direct access
to the
globally
best
matching
category
well let's
say a little
more about
vigilance
vigilance
determines
what features
are learned
in the
critical
feature
pattern
it
clarifies
how our
brains
learn
concrete
knowledge
for some
tests
and abstract
knowledge
for others

in particular
high vigilance
leads to
learning of
narrow
concrete
categories
like a
category
that fires
selectively
to a frontal
view of your
mother's
face
low
vigilance
leads to
learning
of
broad
and
abstract
categories
like
everyone
has
a
face
it
should
be
emphasized
that
critical
feature
patterns
are

explainable
at
every
level
of
vigilance
!
It's
known
from
physiological
experiments
by Desimone
again
that
their
vigilance
control
in the
infratemporal
cortex
which they
showed
by studying
easy
versus
difficult
discriminations
in monkeys
and
in the
difficult
condition
which you'd
assume
would give
you high
vigilance

as expected
you had
enhancement
of the
responses
and
sharp
and seal
activity
to the
attended
stimuli
how is
vigilance
computed
well let's
say of
input
vector
it
instates
a vector
of
activities
and feature
detectors
at the
same time
as it
tries to
activate
the earning
system
but it
does so
multiply
by a
parameter

rho
which is
a sensitivity
or gain
parameter
that's
vigilance
and as
these features
get in state
they try to
shut off
the orienting
system
and if
the excitation
is less
than the
inhibition
the orienting
system
stays quiet
so the
system can
resonate
and learn
but if
inhibition
isn't strong
enough
the orienting
system
gets activated
you get
reset
and search
for new
categories

there's a
very simple
computation
because
you have
an orienting
system
that's
complementary
to the
attentional
system
well how do
you change
vigilance
based on
predictive
success
for this
we have
to go
from
unsupervised
to supervised
art models
so we'll
have
an unsupervised
art a model
an unsupervised
art b model
linked together
by a learned
associative map
as occurs
in fuzzy
art map
and a key

point is
you can
have an input
here
that can create
an output
there
because you
have both
bottom up
and top down
connections
at all
these levels
so in this
way you can
learn
many to
one and
one to
many maps
one example
of a many to
one map
is let's say
you're trying
to categorize
visually
processed
a letter
a which
comes in
multiple
fonts
you'll learn
various visual
categories of
a based

on visual
similarity
at the same
time you're
learning
auditory
categories
saying
a and
then the
associative
map can
map all
of these
visual
categories
of different
a's
to saying
a but
it could
have been
here that
these inputs
were symptoms
tests and
treatments in
a medical
database
prediction
example and
you're predicting
length of stay
of the hospital
the possibilities
here are endless
and there's
many applications

or let's
say
you're trying
to figure out
what this
image is and
you've learned
to say
that's a dog
but today
you say
it's rover
and that
causes a
mismatch
which drives
a search
to focus
attention on
the particular
combination of
features
in this dog
that will
identify it
as rover
that leads
to learning
of a visual
category of
rover
an auditory
category for
the name
rover
an associative
map between
them

and you can
now simultaneously
store expert
knowledge about
that image
well how do you
conjointly minimize
predictive error
and maximize
generalization
so that you
minimize
using
memory
resources
let me read
you an answer
and then show
what it means
in images
match tracking
realizes a
minimax
learning principle
namely given
a predictive
error
vigilance
increases just
enough
to trigger
search
and thus
sacrifices
the minimum
generalization
to correct the
error

so let's say
you've made
a prediction
that must mean
that vigilance
is less
than the analog
match between
bottom up
and top down
but let's say
now you
have a mismatch
well that'll
lead to a
match tracking
signal
that bumps
vigilance up
till it's just
above the analog
match
just big enough
to drive
the search
so you've given
up the minimum
amount of
generalization
to correct
the error
well our
mechanisms
like vigilance
control
realized
in lamina
cortical

and thalamic
circuits
the answer
is yes
my PhD
student
Max Versace
and I
showed this
by developing
the synchronous
matching art
or smart
model
which
introduced
a lot
more
neurophysiological
and anatomical
versimilitude
into the model
including
spiking
dynamics
lamina
cortical
circuits
interacting
with specific
and non-specific
thalamic nuclei
this is another
example of
lamina computing
and here's a
schematic
of the model

you see all
the cortical
layers with
identified cells
a hierarchy
of cortical
regions
interacting with
specific
thalamic nuclei
and non-specific
thalamic nuclei
a ton
of anatomical
data got
functionally
explained in
this way
and many
other data
as well
for example
we showed
if you have
a good enough
match between
bottom up
and top down
you're going
to get
fast gamma
oscillations
during a
tension
there was
quite a bit
of data
about that

already
but we also
showed if you
have a big
enough mismatch
you'll get
slower beta
oscillations
that wasn't
well known
but since
that time
there have
been experiments
in at least
four labs
in three
different parts
of the brain
confirming
that prediction
most important
vigilance
control
was shown
how to be
controlled
by mismatch
mediated
acetylcholine
release
a big enough
mismatch
in the
non-specific
thalamic
nucleus
activates

nucleus
bacillus
of minor
that
releases
acetylcholine
and layer
five cells
across the
cortex
reducing
after
hyperpolarization
currents
and causing
vigilance
to go up
and I also
showed that
breakdowns
in acetylcholine
modulation
can help
to explain
the symptoms
of multiple
mental disorders
so as to
memory
consolidation
we know
there's a
dynamic phase
of memory
consolidation
while the
input
exemplar

still drives
memory
search
and before
direct access
occurs
but what if
the orienting
systems cut out
what if you
have a lesion
in the hippocampus
well then
as occurs
in medial
temporal
amnesia
you get
unlimited
anterograde
amnesia
because you
can't search
for new categories
you get
limited
retrograde
amnesia
because you
can have
direct access
to previously
learned categories
this is a failure
of consolidation
which is
mediated
by the

orienting
system
you get
defective
novelty
reactions
because that
is also
mediated
by the
orienting
system
and memory
consolidation
novelty
detection
are mediated
by the
same
structures
for the
same
reason
there's
normal
priming
because
priming
occurs
within the
attentional
system
learning
of the
first
item
dominates
you

can't
get
some
learning
but you
can't
then
search
and there's
an impaired
ability
to attend
to relevant
dimensions
of stimuli
again
because you
can't
search
so now
where does
infratemporal
cortex fit
in within
the larger
brain
I
introduced
the
predictive
art
or part
algorithm
model
in order
to show
how the
prefrontal

cortex
among
other
things
learns
to
control
all
higher
order
intelligence
you can
find that
in a
2018
paper
on my
webpage
I also
published
it
open
access
and in
this
macro
circuit
these
green
areas
of
prefrontal
cortex
control
processes
like
working
memory

learned
plans
prediction
optimized
action
these
regions
in red
control
processes
like
reinforcement
learning
emotion
motivation
adaptively
timed
learning
the
category
learning
I've
talked
about
in
IT
is
just
in
those
two
regions
all
these
processes
control
visual
perception

and there
are
detailed
models
of all
of
these
regions
and
their
interactions
now
and each
brain
region
in
nature
and
in
predictive
art
carries
out
a
different
function
contrasting
really
dramatically
with the
homogeneous
organization
of a
typical
deep
learning
network
so

I've
told you
just a
little
bit
about
some
aspects
of
cognition
and
why
they're
explainable
but if
you put
in
all the
biological
models
of
perceptual
cognition
emotion
and
action
they're
all
explainable
and
then
you
can
assert
how
perceptual
and
cognitive

processes
use
art
like
excitatory
matching
and
match
based
learning
to
create
self
stabilizing
attentive
and
conscious
representations
of
objects
and
events
that
embody
increasing
expertise
about
the
world
moreover
complementary
spatial
and
motor
processes
that I
couldn't
mention

at all
use
inhibitory
matching
and
mismatch
based
learning
to
continually
update
spatial
and
motor
representations

to
compensate
for bodily
changes
throughout
life
taken
together
they
provide
a
self
stabilizing
perceptual
and
cognitive
front
end
for
conscious
awareness
and

knowledge
acquisition
which
can
intelligently
manipulate
the more
labile
spatial
and
motor
processes
that
enable
our
changing
bodies
to
act
effectively
on
a
changing
world
and
when
you
put
them
all
together
they
provide
a
blueprint
for
designing
autonomous

adaptive
algorithms
and
mobile
robots
with
behaviors
humans
can
understand
and
control
because
they're
both
explainable
and
reliable
see
my
web
page
sites.
bu.
edu
Steve
G.
for
these
models
and
with
that
I'd
like
to
thank
you

very
much
for
your
attention.

Before
we get
started
what I
want
to say
so
I
should
say
that
everything
I've
talked
about
and
much
more
is
in
my
book
conscious
mind
resonant
brain
how
each
brain
makes
a
mind
for

those
who
don't
know
it's
self
contained
and
non
technical
it's
written
in a
conversational
style
so that
people
who
know
nothing
about
the
mind
or
the
brain
can
enjoy
reading
it
and
I
have
friends
who
are
a
rabbi

a
minister
a
painter
a
gallery
owner
a
lawyer
a
social
worker
who've
all
been
enjoying
reading
it
also
it's
a
big
book
it's
almost
800
double
column
pages
with
over
600
color
figures
so
everything
is
illustrated

but
instead
of
costing
\$150
it
costs
\$35
for
the
hard
copy
and
only
\$17
for
the
kindle
because
I
spent
a lot
of
my
own
money
so
that
people
who
are
interested
in
the
topic
can
read
it

and
one
other
comment
if
people
do
have
questions
or
comments
about
my
lecture
or
anything
they
read
in
the
book
my
email
is
just
steve
at
bu
austin
university
dot
edu
and
I'll
be
happy
to
try

to
reply
so
thank
you
now
some
researchers
in
explainable
AI
people
like
Lina
and
Antonio
Di
Cecho
demand
that
any
explainable
AI
should
at the
very
least
meet
these
four
criteria
to
be
fair
not
biased
in
one

way
or
another
to
be
accountable
or
reliable
to
be
secure
against
malicious
hacker
attacks
and
not
to
be
fooled
easily
and
also
to
be
transparent
now
you
explained
how
adaptive
resonance
theory
or
art
and
by the
way

I gotta
say
I love
your
creative
and clever
use of
acronyms
for your
models
my favorite
one is
sovereign
model
self-organizing
vision
expectation
recognition
emotion
intelligent
goal-oriented
navigation
if I'm
correct
amazing
anyway
you explained
how art
can address
and overcome
the issues
of
accountability
security
and
transparency
of
current

deep
learning
approaches
but
it
seems
that
this
fairness
issue
aka
the
problem
of
algorithmic
bias
has
also
been
a
growing
concern
lately
especially
since
it's
regarded
by
some
researchers
like
Antonio
Barria
as a
practically
intractable
problem
so

I wanted
to ask
in what
ways
do you
think
art
can
contribute
to
the
ongoing
quest
for
mitigating
this
problem
well
when
Ali
sent me
this
question
I said
well
first
I'd like
you to
send me
a
definition
of
algorithmic
bias
that
will
clarify
what

you
have
in
mind
so
that
I
know
what
I'm
trying
to
respond
to
and
you
wrote
me
that
you
borrowed
the
term
from
body
as
book
the
information
manifold
and
you
sent
me
a
quote
from
page

247

that

I

will

quote

in

part

before

I

respond

to

that

background

information

so

there

are

two

main

reasons

for

algorithmic

approach

to

decision

making

that

may

result

in

unfair

outcomes

either

at

the

individual

or

group

level
one
is
that
data
used
is
biased
and
another
is
that
the
algorithm
analyzes
the
data
in
such
a
way
that
it
yields
biased
results
the
basic
point
to
remember
is
that
algorithms
are
designed
to
achieve

a
certain
goal
not
created
naturally
by
evolution
or
accident
thus
most
algorithms
are
written
to
detect
certain
patterns
of
interest
for
a
particular
objective
not
just
any
pattern
to
be
able
to
pick
out
some
patterns
and

disregard
others
programmers
build
a
model
of
the
data
by
listing
expectations
about
what
data
should
be
like
in
order
to
qualify
as
relevant
to
the
problem
well
as
I'll
explain
below
self
organizing
learning
classification
prediction
models

like
adaptive
resonance
theory
or art
overcome
all the
problems
it's a
general
purpose
device
but why
don't
I
try
to
answer
that
as
part
of
my
replies
to
all
these
subsequent
questions
okay
thank you
so
much
now
as you
also
mentioned
in your

lecture
in
1988
you pointed
out
17
issues
with
back
propagation
in
one
of
your
most
famous
and
highly
cited
papers
on
nonlinear
neural
network
so
it's
been
34
years
now
do
you
see
any
fundamental
Copernican
change
of

perspective
happening
in
deep
learning
research
or
we
still
keep
adding
epicycle
upon
epicycle
to our
Ptolemaic
model
well
you've
sort
of
anticipated
what
I'm
going
to
say
and
as
I
said
in
my
lecture
various
investigators
and
I

mentioned
Kuhn
Kirkpatrick
and
Veles
have
recently
attempted
to
modify
deep
learning
to
overcome
some
of
its
problems
but
as
Ali
just
mentioned
my
lecture
noted
that
at least
to
my
mind
they're
like
epicycles
that
are
added
to

a
kind
of
Ptolemaic
model
of
the
solar
system
to
overcome
some
of
its
problems
but
as
we
all
know
the
Ptolemaic
model
ultimately
crashed
because
it
was
both
qualitatively
and
quantitatively
wrong
and
they
could
only
be

solved
by
throwing
out
the
Ptolemaic
model
and
replacing
it
with
the
Copernican
model
that
became
the
basis
for
modern
astronomy
and
astrophysics
so
art
overcomes
foundational
deep
learning
problems
that
can't
be
solved
using
epicycles
and
it

already
done
it
shortly
after
I
introduced
it
in
1976
and
it
can't
be
overemphasized
as I
noted
in my
lecture
deep
learning
is
untrustworthy
because
it's
not
explainable
and
it's
unreliable
because
it can
experience
catastrophic
forgetting
and
that
happens

for
basic
reason
deep
learning
just
like
back
prop
which
is
its
learning
engine
is
just
a
feed
forward
adaptive
filter
so
as you
note
in your
question
I
described
these
two
problems
in
addition
to
15
others
in
my

oft
cited
article
that
I
published
in
1988
in
the
first
issue
of
neural
networks
and
I
also
showed
them
that
art
had
already
solved
the
problems
in
1976
and
what
I
find
sad
is
that
back
propagation

and
deep
learning
architects
like
Jeff
Hinton
who
knows
all
of
this
background
never
mention
this
history
and
keep
talking
about
making
deep
learning
explain
the
brain
but
it
can't
explain
the
brain
because
its
foundation
is
contradicted

by
basic
psychological
and
neural
data
yes
someone
should
great
in the
deep
learning
community
I
like
comparative
!
discussion
and
criticism
but I
don't
like
solipsism
in
science
great
thank
you
now
on
slide
number
50
of your
presentation
you pointed

out that
art
is
inconsistent
with models
where
top-down
matches
suppressive
such as
Bayesian
explaining
away
a similar
view is
evident
on page
195
of
conscious
mind
resonant
brain
to which
you also
add
one
of
many
serious
problems
of
the
Bayesian
models
is
that
fully

suppressive

matching

circuits

cannot

solve

the

stability

plasticity

dilemma

now

would you

care

to

further

elaborate

on

this

point

sure

my

lecture

and

my

book

summarizes

my

book

my

lecture

couldn't

go

into

a lot

of

it

some

of

the

copious
psychological
anatomical
and
neurophysiological
evidence
expectations
which
obey
what
Gail
Carpenter
and I
call
the
art
matching
rule
are
matched
against
bottom
up
input
patterns
and
as
the
lecture
briefly
noted
the
art
matching
rule
is
defined
by

a
modulatory
on
center
off
surround
network
and
the
modulatory
on
center
is
excitatory
however
acting
by
itself
it
can't
fully
excite
its
target
cells
it
can
prime
them
sensitize
excuse
me
or
modulate
them
to be
ready
to fire

vigorously
when
matched
bottom
up
inputs
arrive
and
when
there
is
a
good
enough
match
between
the
bottom
up
inputs
and
an
active
top
down
expectation
that's
reading
out
a
circuit
that
obeys
the
art
matching
rule
that's

when
you
get
what
I
noted
in
my
lecture
what
I
call
a
feature
category
resonance
because
it
develops
between
the
matched
or
attended
features
and
the
recognition
category
that
they
activate
and
it's
this
resonance
that
synchronizes

and gain
amplifies
the
matched
features
well
suppressing
the
mismatched
features
and
that
sustained
resonance
is
important
because
it's
sustained
long
enough
to
drive
learning
in
the
more
slowly
varying
adaptive
weights
of the
active
bottom
up
filter
and
learn

top
down
expectation
and
it's
because
resonance
triggers
learning
that
I
call
the
theory
adaptive
resonance
theory
and
the
art
matching
rule
avoids
catastrophic
forgetting
as I
briefly
mentioned
in the
lecture
because
it
suppresses
irrelevant
features
using
its
off

surround
while
it's
amplifying
and
focusing
attention
on the
critical
features
that
regulate
both
bottom
up
and
top
down
learning
as
well
as
successful
predictions
because
they're
relevant
they've
been
selected
by
previous
learning
experiences
which
discover
the
set

of
features
that
are
predictive
or
causal
in a
given
situation
and
along
the
way
not only
does
the
art
matching
rule
achieve
causality
in predictions
although
as the world
changes
you have
to
update
your
causal
explanations
it
also
solves
the
stability
of

plasticity
dilemma
in
brief
purely
suppressive
matching
can't
do
any
of
this
it
shuts
off
the
expected
data
and
so
it
can't
focus
attention
or
learn
about
it
and
there
is
fully
suppressive
matching
in
spatial
and
modal

learning
but
that
isn't
learning
to be
expert
about
the
world
I
could
explain
that
more
if you
want
to
know
but
that's
also
in
my
book
and
these
two
kinds
of
learning
the
excitatory
match
based
learning
and
the

inhibitory
dispatch
learning
are
computation
complementary
it's
another
example
of
complementary
computing
and
the
match
based
learning
goes
on
in
the
ventral
or
watt
cortical
stream
and
the
mismatch
learning
goes
on
in
the
where
or
dorsal
cortical

stream
the
what
stream
for
perception
and
categorization
and
prediction
the
where
stream
for
spatial
representation
and
action
and
then
you
need
what
to
where
and
where
to
what
interactions
so that
you
can
reach
for
and
otherwise
engage

to
approach
and
what
have
you
look
at
reach
for
approach
to
things
that
you
recognize
thank you
now
following
from the
previous
question
and
considering
that
the
free
energy
principle
and
active
inference
framework
as
works
in
progress
are

related
to
predictive
coding
and
bayesian
brain
hypothesis
what
is
your
view
on
the
extent
of
compatibility
between
art
and
active
inference
because
despite
some
primofacial
similarities
between
the
two
do
you
see
them
as
fundamentally
incompatible
or

irreconcilable
and
how
could
this
issue
be
rigorously
evaluated
and
positively
resolved
in terms
of
reconciliation
or
integration
of
art
and
active
inference
or
otherwise
because
you
see
to add
some
more
context
here
Smith
et al
in
their
recent
paper

an
active
inference
approach
to
modeling
structure
learning
have
stated
that
although
they
have not
explicitly
incorporated
arts
top-down
attentional
and feedback
mechanisms
there are
mechanisms
within
their
active
inference
based
model
which
they
believe
are
quite
similar
to
top-down
and

bottom-up
feedback
exchange
in
art
so
there
seems
to be
some
degree
of
disagreement
about
the
compatibility
between
the
two
frameworks
well
let me
try
to
respond
to
the
two
parts
of
your
question
separately
so
I'm
not
going
to

try
to
talk
about
Smith
at all
for
a
moment
let's
talk
about
free
energy
and
I
like
getting
definitions
on
the
table
because
it's
really
so
frustrating
to
try
to
remember
what
something
is
when
someone
is
talking

about

it

so

I

go

to

Wikipedia

Wikipedia

writes

in part

that

the

free

energy

principle

assert

quote

that

systems

minimize

a free

energy

function

of their

internal

state

which

entail

beliefs

about

hidden

states

in

their

environment

the

implicit

minimization

of free
energy
is
formally
related
to
variational
Bayesian
methods
and
was
originally
introduced
by
Carl
Friston
as
an
explanation
for
embodied
perception
neuroscience
where
it's
also
known
as
active
inference
and
we
all
know
Carl
is
a
brilliant

and
very
insightful
man
the
free
energy
principle
describes
the
behavior
of a
given
system
by
modeling
it
through
a
Markov
blanket
that
tries
to
minimize
the
difference
between
their
model
of the
world
and
their
sense
and
associative
perception

this
difference
can be
described
as
surprised
and
is
minimized
by
continuous
correction
of the
world
model
of the
system
one
more
part
of the
quote
the
free
energy
principle
has
been
criticized
for
being
very
difficult
to
understand
each
and
even

for
experts
and
the
mathematical
consistency
of the
theory
may have
been
questioned
by
recent
studies
discussions
of the
principle
have
also
been
criticized
for
invoking
metaphysical
assumptions
far removed
from
attestable
scientific
prediction
making
the
principle
unfalsefiable
and
in a
2018
interview

Fristen
acknowledged
that
the
free
energy
principle
is
not
properly
fortifiable
so
that's
Fristen
himself
so
my
main
concern
with
the
free
energy
principle
just
like
any
theory
about
how
brain
makes
a
mind
is
how
much
data

can
it
explain
in
a
principle
and
unifying
way
that's
what
we
do
in
science
we
develop
theories
to
explain
and predict
data
and
in
the
case
of
the
free
energy
principle
from
what
I
can
see
the
answer

is
essentially
no
data
and
you
can
correct
me
if
I'm
wrong
it
therefore
cannot
be
evaluated
as a
physical
theory
at all
and
there's
a
basic
reason
for
this
problem
our
brains
are
designed
to
autonomously
learn
in
real

time
in
response
to
a
changing
or
non
stationary
world
that's
filled
with
unexpected
events
like
today
we're
experiencing
an
unexpected
event
I
didn't
know
until
recently
that
I'd
be
enjoying
your
company
today
optimization
principles
were designed
to cope

with
stationary
dynamics
whose
rules
and
probabilities
do
not
change
through
time
so
it's
not
possible
to
quote
minimize
the
difference
between
their
model
of
the
world
and
their
sense
and
associated
perception
unquote
because
there
is
no

predefined
model
of
the
world
which
is
always
changing
in
unexpected
ways
so
you
need
a
theory
about
how
the
world
changes
surprise
occurs
in
art
when
there's
a
big
enough
mismatch
between
an
input
pattern
and
the

currently
active
top
down
expectation
of
a
category
that
it's
activating
this
mismatch
activates
the
art
orienting
system
that
I
briefly
discuss
in
my
talk
which
interacts
with
the
attentional
system
where
the
category
learning
doesn't
occur
and

as
I
illustrated
in
our
discussions
of
search
and
vigilance
that
it
drives
hypothesis
tests
in
a
memory
search
to
discover
a
better
match
or
to
begin
to
learn
new
category
so
art
will
discuss
a
better
match

in
the
case
where
the
system
was
attending
some
other
familiar
features
when
the
new
input
occurs
but
the
features
of
the
new
input
have
previously
been
categorized
that's
why
they're
familiar
and
then
the
orienting
system
very

quickly
shifts
attention
to
the
matching
category
and
you
resonate
on
and
you
recognize
it
consciously
often
art
begins
to
learn
a
new
category
when
the
input
represents
a
truly
unfamiliar
and
novel
situation
now
as to
Bayesian
methods

in
science
hey
I'm a
mathematician
how can
I not
love
Bayes
right
but the
beauty
of Bayes
is its
simplicity
you just
write the
probability
of two
events
a and
b
in two
different
ways
the
probability
of
b
given
a
times
the
probability
of
a
the
probability

of
a
given
b
times
the
probability
of
b
set
them
equal
because
they're
identical
divide
by
let's
say
probability
of a
and
then
optimize
that's
Bayes
and
it's
a useful
statistical
method
and should
continue
to be
used
in
statistics
but

it's
just
a
formal
identity
wherein
lies
its
power
it
says
nothing
about
any
physical
reality
whether
in
physics
chemistry
or
biology
the
Bayes
rule
itself
tells
us
nothing
about
physical
reality
and
contains
no
heuristics
to
discover

anything
about
physical
reality
but
that
you
need
to
develop
models
driven
by
a
profound
analysis
of
large
databases
so
it
turns
out
that
biological
models
like
art
do
not
incorporate
the
Bayes
rule
however
art
does
routinely

choose
the best
or
optimal
categories
that
represent
the
data
best
so
you
don't
need
Bayes
to
achieve
optimality
also
Bayes
works
best
in a
stationary
world
with
stationary
probabilities
and
art
designed
to learn
about
a
non-stationate
world
so
you

know
one
can
discuss
this
until
the
cows
come
home
it's
good
for
what
it
was
designed
for
and
some
of
the
neuroscientists
who
try
to
apply
Bayes
are
wonderful
experimentalists
but
they
know
no
math
and
no

theory
and
you
know
it's
the
temptation
of
free
lunch
there
it
is
waiting
to
be
applied
there
is
no
free
lunch
in
science
thank
you
now
as a
final
point
of
comparison
what
are
the
Smith
I
didn't

reply
to
Smith
yes
yes
okay
you
quoted
a
sentence
of
Smith
but
before
that
said
Smith
et
all
wrote
it
is
also
worth
highlighting
that
as
our
model
is
intended
primarily
as
a
proof
of
concept
and

a
demonstration
of
an
available
model
expansion
reduction
approach
that
can
be
used
within
active
inference
research
it
does
not
explicitly
incorporate
some
aspects
such
as
top
down
attention
that
are
of
clear
importance
to
cognitive
learning
processes

and
that
have
been
implemented
in
previous
models
for
example
the
adaptive
resonance
theory
of
growth
was
designed
to
incorporate
top

down
attention
mechanisms
and
feedback
mechanisms
to
address
a
fundamental
knowledge
accusation
problem
the
temporal
instability

of
previously
learned
information
that
can
occur
when
a
system
also
remains
sufficiently
plastic
to learn
new
and
potentially
overlapping
information
while
our
simulations
do not
explicitly
incorporate
these
additional
complexities
there
are
clear
analog
to
the
top
down
and

bottom
up
feedback
exchanges
within
our
model
such
as
the
prediction
and
prediction
error
signaling
within
the
neural
process
theory
associated
with
active
inference
or
addresses
the
temporal
instability
problem
primarily
through
mechanisms
that
learn
top
down
expectancies

that
guide
attention
and
match
them
on
bottom
up
input
patterns
which
is
quite
similar
to
the
prior
expectations
and
likelihood
mappings
used
within
active
inference
but
as
I've
already
noted
the
quote
prior
expectations
and
likelihood
mappings

within
adaptive
inference
quote
unquote
do not
have
any
of
the
key
learning
attention
and
memory
stability
properties
of
the
york
matching
rule
the
york
matching
rule
is
unique
solution
to
that
problem
in
its
variations
it's
been
supported

by
psychological
anatomical
physiological
and
biophysical
data
it also
occurs
in many
species
nobuasuga
for example
shows
it occurs
in
bats
it
occurs
in
ferrets
you
know
so
also
I think
it's
important
to
know
that
when
learning
begins
in
an
art
model

it
doesn't
need
prior
expectations
or
likelihood
in
fact
typically
the
initial
bottom-up
weights
are chosen
to be
random
because you
don't know
what you're
going to be
experiencing
and the
initial
top-down
expectations
are chosen
to be
large
so that
whatever
category
happens
to be
learned
when it
reads
out

its
top-down
expectation
it
can
match
whatever
features
activated
that
category
so
they
all
start
larger
and
then
they're
pruned
to match
the
critical
features
that
happen
to be
learned
in
that
category
so
there
are
no
built-in
models
art

discovers
its
own
models
I
should
also
emphasize
that
active
inference
is
also
not
explainable
arts
is
explainable
because
a
currently
active
critical
feature
activity
pattern
namely
the
features
to
which
attention
is
paid
controls
all
learning
and

prediction
by
the
model
and
in
principle
can
be
measured
by
neurophysiological
experiment
a
model
without
cell
activities
or
short-term
memory
traces
that
can
represent
the
critical
feature
pattern
can't
be
explainable
so
I
think
there
are
qualitative

differences

I

don't

say

people

shouldn't

use

active

inference

it

may

be

incredibly

useful

and

powerful

in

technological

application

but

when

one

is

doing

you

know

brain

science

psychology

it

just

doesn't

match

foundational

data

things

thanks

and

I guess
you
somehow
already
answered
a part
of this
question
but
what are
the possible
ways
in which
arts
approach
to
explainable
AI
which
if I'm
not
mistaken
can be
described
as a
model
dependent
intrinsically
explainable
approach
can inform
active
inference
approach
and cross
fertilize
with it
which is

based on
abductive
reasoning
through
constructing
generative
models
for example
as is
sketched out
in
Pahr and
Pizzullo's
understanding
explanation
and active
inference
paper
well
first
I don't
think
art
is
model
dependent
as I
just
noted
one
begins
typically
to learn
in art
with
random
initial
bottom

up
weights
and
uniformly
distributed
dot down
initial
adaptive
weights
so you
can
match
any
category
that
you
happen
to
learn
but
the
authors
you
quoted
write
in part
that
active
inference
and here
I want
to
quote
them
so
I can
respond
in a

little
more
detail
active
entrance
quote
implies
a deep
generative
model
that
includes
a model
of the
world
used to
infer
policies
and a
higher level
model
that
attempts
to predict
which
policies
will be
selected
based upon
a space
of hypothetical
that is
down
of factual
explanations
and which
can
subsequently

be used
to provide
retrospective
explanations
about the
policies
pursued
so
again
art works
without a
generative
model of
the world
or any
predefined
!
policies
because what
it's trying
to discover
what changing
world it
happens to be
in and nobody
knows what
it is
a priori
and in
general
an art
classifier
responds
to a
front
end
of
preprocesses

that
process
perceptual
data
from one
or another
sense
notably
vision
and audition
where we
get
most of
our
information
about the
world
and that's
why a
classifier
like art
begins
its work
in the
brain
in the
temporal
cortex
where it
receives
highly
preprocessed
perceptual
representation
so
decades
of work
went into

understanding

how a

brain is

consciously

seen

in the

ear

and in

the case

of

vision

art

classifies

perceptual

boundaries

and

surfaces

that

require

multiple

stages

of

processing

because

as I

mentioned

briefly

they

are the

outcome

of what

I call

hierarchical

resolution

of

uncertainty

you need

multiple

stages
to
define
a
perceptual
boundary
surface
one reason
being
because
our
sensory
organs
register
such
noisy
and
incomplete
data
like
you may
know
that
our
photosensitive
retina
has a
huge
blind
spot
where
you
can't
register
any
sense
any
visual

signal
the
blind
spot
as big
as
the
fovea
where
all of
our
high
resolution
vision
occurs
so
it's
not a
little
thing
and
moreover
veins
come out
of the
fovea
and
occlude
the
retina
in
multiple
places
and
you
can't
register
visual

signals
on the
veins
either
so
the
signal
you're
getting
is
very
incomplete
and
it
takes
multiple
processing
stages
to
overcome
those
uncertainties
and
my
colleagues
and I
have
worked
for
decades
to
explain
how
that
happened
maybe
I'll
stop

there
for
that
thank
you
so
much
now
this
next
question
is of
a
personal
interest
to me
because
currently
I'm
working
on
modeling
some
probabilistic
aspects
of
affective
response
to
music
and
your
most
recent
paper
toward
understanding
the

brain
dynamics
of
music
immensely
helped
me
gain
a
better
understanding
of
entrainment
as
you
pointed
out
in
the
supplementary
notes
for
this
paper
violation
of
prior
learned
expectations
is
instrumental
in
inducing
a
wide
range
of
affective

responses
in
musical
and
non-musical
situations
some
psychologists
such as
Patrick
Uslin
have
distinguished
between
perception
and
arousal
of
emotions
but you
left
out
a
question
is
it
a
lack
of
time
or
you
just
skipped
it
I
want
quite

a
bit
about
it
how
do
you
see
the
future
of
art
and
neuro
inspired
AI
research
I
think
that
will
be
our
last
question
okay
so
we'll
come
back
to
that
yes
yes
thank
you
because
that

that's
an
important
question
yes
okay
so
sorry
to
interrupt
I
just
want
to
be
sure
no
problem
thank
you
now
yes
well
some
psychologists
such as
Patrick
Uslin
have
distinguished
between
the
perception
and the
arousal
of emotions
in the
context

of
musical
experience
and
also
several
studies
such as
the works
of Uslin
and
Gabrielson
from
the
psychology
department
of
Uppsala
University
have
shown
that
despite
music's
ability
to
communicate
a
wide
range
of
positively
and
negatively
valenced
emotions
it
somehow

evokes
mostly
positively
valenced
emotions
for instance
we can
easily
perceive
rage
or anger
in music
without
necessarily
getting
angry
on the
other
hand
we're
more
likely
to
actually
feel
elevated
and
happy
after
listening
to
happy
music
and
also
the
evidence
shows

that
this
disparity
between
perception
and
evocation
of
emotion
is
probably
even
more
significant
in
musical
experiences
than
any
other
non-musical
experiences
so
how can
this
difference
in
diversity
between
perceived
and
aroused
or evoked
musical
emotions
be accounted
for
within

art
framework
can it
possibly
be regarded
as
another
kind
of
broken
symmetry
as you
mentioned
on page
621
of your
book
but
specific
to
musical
effects
so
as
you've
noted
I
haven't
studied
these
issues
in the
context
of
music
but
I'll
try

to
venture
some
general
comments
I should
first
note
that
the
laminarch
model
which
is
a
development
of
art
to
show
how
and
why
all
neocortical
circuits
that
support
perception
and
cognition
typically
share
canonical
six
layer
circuit
my

colleagues
and I
have
modeled
how
just
as
in our
brains
variations
of
this
canonical
laminar
circuit
can
support
all
perceptual
and
cognitive
processes
so
there's
a
major
generalization
of
art
and
we've
done
it
for
vision
speech
and
cognitive

working
memory
and
planning
in
particular
so
the
main
point
is
that
the
laminar
circuitry
is
basically
in all
perceptual
and
cognitive
areas
vision
auditions
that
that's
how
one
can
create
a
context
for
discussing
music
and
in
fact

my
work
on
music
applies
such
discoveries
I
was
able
to
put
together
discoveries
that
had
been
made
based
on
what I
believe
were
different
evolutionary
pressures
on
the
organization
of our
brains
but that
evolution
also
discovered
and I
try to
sketch

how
if you
put
some
of them
together
in a
certain
way
then
capacity
for
learning
and
consciously
performing
music
could
arise
so now
how about
arousal
well
it's
essential
of course
to all
awareness
and
consciousness
your
cortex
needs
to be
adequately
aroused
for waking
consciousness

to occur
at all
it
also
arousal
plays
a major
role
in the
processing
of emotions
and is
very relevant
to musical
issues
because my
gated
dipole
model
explains
how
opponent
processes
opposites
are organized
in all
parts of
our brain
perceptual
cognitive
motor
affective
in particular
emotions
are organized
in pairs
in such
an emotional

dipole
and one
reason is
because emotions
need to compete
with each other
such as fear
versus
relief
for example
in post-traumatic
stress disorder
therapy
a therapist
may try
to help
a patient
to think
about
positive
experiences
that generate
relief
in order
to inhibit
the chronic
fear
that's so
destabilizing
during PTSD
so their
opposites
are competing
and another
property that
arousal
enables
is that

the sudden
offset
of an emotion
like fear
let's say
during escape
behavior
let's say
you know
a favorite
example
of mine
is
you know
some
cruel
experimentalist
puts the
pigeon
in a
skinner
box
the floor
is electrified
the pigeon
is feeling
pain
and fear
it's
dashing
frantically
around
trying to
keep its
feet off
the floor
it bangs
into a red

buzzer
the buzzer
shuts
the shock
off
and
the animal
experienced
a wave
of relief
or positive
motivation
for learning
the escape
response
so
the rebound
from
theater relief
that can
be associated
with actions
that lead
to escape
and can
motivate escape
by means
in the future
is energized
by arousal
in the gated
dipole
you shut
off the
external
cue
of shock
shock

but the
arousal
is tonically
or
has sustained
activity
in both
the fear
and relief
channels
so
because the
arousal
is sustained
or tonic
through time
and equally
activates
the fear
and relief
channels
when fear
suddenly
decreases
then arousal
in the relief
channel
wins the
competition
and can
thereby call
what I call
an antagonistic
rebound
from theater
relief
that activates
the relief

channel
and thereby
providing
motivation
for escape
whether
reactive
or learned
through escape
experiences
and I also
approve
which is
related
to some
degree
to music
that
the
non-occurrence
of an
expected
event
can by
itself
cause
a burst
of arousal
and thus
an antagonistic
rebound
and can
flip emotions
from positive
to negative
in so doing
and I always
loved that

discovery
because
I especially
love discoveries
where
the occurrence
of nothing
has profound
effects
on future
behavior
so it's
the non-occurrence
of the expectation
cause of this
mismatch
that can
flip
emotions
now how
this influence
of the perception
of music
needs more work
it's
as I mentioned
in my paper
I haven't
tried to
study
that
my paper
on music
I feel
is like
you know
a drop
in the bucket

and hopefully
if I don't
get around
to it
someone else
will
but the above
examples show
that arousal
and emotion
are not
the same
thing
cause arousal
can support
all emotion
fear
relief
hunger
society
whatever
the ones
that win
the competition
are then
able
to support
compatible
behaviors
by motivating
them
and I've
also explained
that the level
of arousal
must be chosen
within an
intermediate

range
to support
normal behaviors
it's a kind
of golden
mean
there's an
inverted
U
on the
effects
of having
arousal
from too
little
or too
much
and if
you have
too little
arousal
you have
an under
aroused
syndrome
which can
support
symptoms
like
autism
and
over
arousal
can support
symptoms
like
schizophrenia
is only

one of
many
factors
in
these
diseases
but
I'm
happy
to say
that
subsequent
clinical
data
supported
those
predictions
that
these
two
mental
disorders
are
at
the
opposite
speed
of
the
arousal
inverted
U
so
I
think
and
you know
it's

very
speculative
because
I've
never
really
seriously
studied
it
and
I
try
not
to
speculate
but
what
the
hell
the
kind
of
arousal
that
music
activates
is
generally
positive
just
like
the
arousal
that
activates
exploratory
behaviors
is

positive
there's
somehow
linked
you know
music
is a
sonic
adventure
if you
like
there's
no
aversive
cue
when
listening
to
music
except
perhaps
music
that
played
so
loud
as
to
cause
a
headache
or
ear
damage
or
even
a
seizure

in
susceptible
individual
stuff
there's
no
particular
reason
why
it
shouldn't
be
positive
so now
which
question
you
want
to
ask
me
I
think
we're
left
with
just
two
other
questions
if
not
too
tired
starts
with
as
a

final
point
of
comparison
another
starts
on
a
more
further
side
yes
the
last
where
and
the
last
two
questions
so
you
want
to
how
do
you
see
the
future
of
all
immune
spider
air
research
and
before

that
yes
before
that
I
just
wanted
to
ask
you
about
the
your
view
about
artificial
consciousness
do
you
see
I
really
need
the
first
answer
to
answer
the
second
question
okay
as
you
wish
yes
okay
so

so
yeah
and
how
do
you
see
yeah
yes
it's
quite
fine
and
how
do
you
see
the
future
of
art
and
brain
inspired
AI
research
in
general
in
your
view
what
research
areas
ought
to
gain
more

attention

than

they

do

today

so

I'll

give

you

quite

a

general

answer

but

it

implies

what

I

think

about

this

so

first

with

the

caveat

I

like

to

say

I

couldn't

predict

the

present

so

I

can't

predict
the
future
that
being
said
I
believe
that
all
engineering
technology
and
AI
will
increasingly
embody
autonomous
adaptive
intelligence
in the
coming
century
and we
can
already
see
its
beginnings
in
autonomous
automobiles
and airplanes
and
increasingly
autonomous
controllers
on the

factory
floor
and many
people
have
written
about
it
and I
think
art
will play
a central
role
in
this
as
well
as
other
models
that
are
summarized
in
my
book
and
that's
because
already
in
1980
I
published
a
thought
experiment

in the
journal
Psychological
Review
which
was
then
and
still
probably
remains
the
leading
theory
journal
in
psychology
and
you
may
recall
that
Einstein
derived
both
special
relativity
theory
and
general
relativity
from
thought
experiments
and
let me
just
clarify

by my
thought
experiment
they
derive
their
enormous
power
so
my
thought
experiment
was
about
how
any
system
and
autonomously
thought
about
autonomy
correct
predictive
errors
in a
changing
world
and
the
hypotheses
upon
which
the
thought
experiment
were
derived

but
just
the
few
facts
that
are
familiar
to
us
all
from
daily
life
and
they're
familiar
because
they
represent
ubiquitous
environmental
pressures
on the
evolution
of our
brains
over
the
millennia
and
when
they
act
together
I
suggest
art

is
the
unique
outcome
that's
a
huge
claim
and
I
turn
to
the
power
of
the
thought
experiment
not
any
personal
ego
trip
for
that
belief
in
particular
nowhere
in the
thought
experiment
of the
words
mind
or
brain
mentioned

so
if
you
accept
that
these
facts
about
the
world
exist
which
we
all
do
and
that
they're
always
operating
on
us
then
you
have
to
accept
the
outcome
if
you
believe
in
the
scientific
method
and
logic

so
art
is
a
universal
solution
to
the
problem
of
autonomous
error
correction
and
changing
world
so
in
another
way
if
you
can't
find
a
mistake
in
the
thought
experiment
then
I
think
you
either
have
to
believe

in
art
like
dynamics
may
be
expressed
in
lamin
art
or
your
favorite
art
variant
or
you
have
to
give
up
your
belief
in
logic
and
the
scientific
method
it
doesn't
imply
that
art
can't
be
further
developed

I
expect
a
large
number
of
scientists
and
technologists
to be
busy
developing
art
like
architecture
long
after
I'm
gone
so
maybe
now
we
can
go
to
your
philosophical
note
with
that
background
yes
thank
you
so
much
now

yes
as I
mentioned
earlier
I
just
wanted
to
ask
about
your
view
about
your
view
on
artificial
consciousness
do
you
see
the
consciousness
as
artificially
producible
or
engineerable
as
some
researchers
like
Mike
Solmsley
is
there
a
fundamental

distinction
between
a
biologically
conscious
agent
and
an
artificial
agent
with
a
fully
simulated
computational
model
of
consciousness
I
know
it's
a
big
big
question
but
I
couldn't
resist
asking
your
opinion
as
an
authority
on
consciousness
modeling

I'm
happy
to
give
a
shot
so
as I
just
noted
my
work
on art
suggests
it
solves
a
universal
problem
about
how we
can learn
to correct
predictive
errors
in a
changing
world
my
work
also
shows
in
its
analysis
of
hierarchical
resolution

of uncertainty
remember
like
how you
go from
some
noisy
retina
to
a
surface
representation
that can
control
looking
and
reaching
how
evolution
may have
been
driven
to discover
conscious
states
so
this
was a
surprise
to me
too
conscious
states
were needed
in order
to choose
that
processing

level
or levels
that
computes
a
sufficiently
complete
context
sensitive
and stable
representation
in the
case of
vision
surface
representation
with which
to
successfully
plan
and act
to realize
value
goals
so
let me
make it
clear
so
you start
with a
noisy
retina
you have
to go
up all
of these
stages

until
you get
a
sufficiently
complete
surface
and boundary
representation
that you
can use
to regulate
successful
action
and if
you used
one of the
earliest
stages
it would
lead to
incorrect
actions
which would
kill you
off by
Darwinian
selection
so how
the hell
do you
know
where the
stage is
where you
compute the
sufficiently
complete
context

sensitive
and stable
one
and I
propose
in vision
I predicted
that the
choice is
embodied in
what I
call a
surface
shroud
resonance
between
pre-strived
visual cortical
area V4
and the
next
processing
stage
posterior
parietal
cortex
or PPC
so it's
in V4
you get
this really
good surface
representation
and then
a resonance
between the
surface
and spatial

attention
which fits
the surface
that spatial
attention
in PPC
is called
a shroud
Christopher
Tyler gave
it that
name
a surface
shroud
resonance
allows you
to pay
conscious
spatial
attention
to
the surface
that you're
going to use
to control
looking and
reaching
behaviors
so it's a
way of ensuring
you have a
good enough
representation
to control
action
so
the shroud
is computed

in posterior
parietal
cortex
which is
part of
the dorsal
or
where
cortical
stream
and
the shroud
modulates
invariant
category
learning
in the
ventral
or
watt
cortical
stream
I can't
go into
that right
now
but my
book
discusses
it
the
category
learning
itself
in the
watt
cortical
stream

as I
indicated
is supported
by
feature
category
resonance
and so
the surface
shroud
resonance
is modulating
in varying
category
learning
in the
feature
category
resonances
surface
shroud
resonance
also supports
conscious
seeing
the feature
category
resonance
is supporting
conscious
recognition
and when
they
synchronize
across
streams
on a
familiar

object
that's
when you
consciously
see
something
that you
know
about
okay
so
conscious
states
hereby
arise
due to
learning
requirements
this
sort of
fell out
of the
wash
of
how you
do
invariant
category
learning
and
learning
in
particular
without
catastrophic
forgetting
it's
regulating

feature
category
resonance
so
given
that the
above
solutions
are
computationally
universal
in the
sense
I
sketch
the
self
organizing
machine
that
embodies
them
should
be able
to support
internal
representations
whose
parametric
!
properties
mimic
conscious
states
whether
such a
machine
can

experience
conscious
qualia
remains as
much of a
mystery
for machines
as it
does
for humans
and
that's
because
no
computational
theory
which
after all
is just
a set
of
equations
can do
more than
imitate
the dynamics
of our
brains
perhaps
with
great
precision
I
don't
have
a
clue
why

the
representations
that
my colleague
and I
have worked
so hard
to explain
huge
amounts
of
psychophysical
data
about
seeing
extra
shading
3D
form
just
go
through
the
list
why
they
support
qualia
I
don't
know
ask
God
or
whatever
God
you
choose

to
believe
in
the
21st
century
thank
you so
much
professor
I
think
we
have
a
couple
of
questions
in
the
chat
but
we
are
actually
approaching
our
two hour
limit
I don't
know
Daniel
if
it's
a good
place
to stop
or

whatever
you say
I think
that's a
great
place
to stop
you've
given us
a lot
to
think about
and digest
and I
hope that
these
words
are
taken
well
and
paid
attention
to
might
create
some
categories
activate
categories
but
professor
Grossberg
thanks
again
for this
amazing
live

stream
we
really
appreciate
it
I
appreciate
it
depending
on
younger
people
like
yourself
to do
just
what
you
said
Daniel
I'm
not
going
to be
around
that
much
longer
so
I
hope
you
have
whether
with my
work
directly
or

related
work
you
have
a
very
fulfilling
intellectual
adventure
I
know
I've
been
on a
wild
ride
since
I
was
17
that's
65
years
of
discovery
and
I
loved
every
minute
of it